M378K: March 6th, 2028.

M378K Introduction to Mathematical Statistics
Problem Set #13
Order Statistics.

Problem 13.1. *An insurance company is handling claims from two categories of drivers: the good drivers and the bad drivers. The waiting time for the first claim from a* **good** *driver is modeled by an exponential random variable T_g with mean 6 (in years). The waiting time for the first claim from a* **bad** *driver is modeled by an exponential random variable T_b with mean 3 (in years). We assume that the random variables T_g and T_b are independent.*

What is the distribution of the waiting time T until the first claim occurs (regardless of the type of driver this claim was filed by)?

Definition 13.1. *Let $Y_1, \ldots, Y_n$ be a* **random sample**. *The random sample ordered in an increasing order is called an* order statistic *and denoted by*

$$Y_{(1)}, Y_{(2)}, \ldots, Y_{(n)}.$$

Question Write $Y_{(1)}$ as a function of $Y_1, Y_2, \ldots, Y_n$.

$$Y_{(1)} = \min(Y_1, \ldots, Y_n)$$

Question Write $Y_{(n)}$ as a function of $Y_1, Y_2, \ldots, Y_n$.

$$Y_{(n)} = \max(Y_1, \ldots, Y_n)$$

Problem 13.1.

$$T = \min(T_g, T_b)$$

$$S_T = [0, \infty)$$

$t > 0$. $F_T(t) = \mathbb{P}[T \leq t]$

$$= \mathbb{P}[\min(T_g, T_b) \leq t]$$

$$= 1 - \mathbb{P}[\min(T_g, T_b) > t]$$

$$= 1 - \mathbb{P}[T_g > t, T_b > t] \quad \text{(independent)}$$

$$= 1 - \mathbb{P}[T_g > t] \cdot \mathbb{P}[T_b > t]$$

$$= 1 - e^{-\frac{t}{\tau_g}} \cdot e^{-\frac{t}{\tau_b}}$$

$$= 1 - e^{-t\left(\frac{1}{\tau_g} + \frac{1}{\tau_b}\right)}$$

$T \sim E(\tau)$ w/ $\tau = \dfrac{1}{\frac{1}{\tau_g} + \frac{1}{\tau_b}} = \dfrac{1}{\frac{1}{6} + \frac{1}{3}}$

$$\Rightarrow \boxed{\tau = 2}$$

$\square$

Problem 13.2. *What is the distribution function of the random variable $Y_{(n)}$?*

$\longrightarrow$: for $y \in \mathbb{R}$: $F_{(n)}(y) = \mathbb{P}[Y_{(n)} \leq y] = \mathbb{P}[\max(Y_1, \dots, Y_n) \leq y]$

$= \mathbb{P}[Y_1 \leq y, Y_2 \leq y, \dots, Y_n \leq y]$ (independence)

$= \mathbb{P}[Y_1 \leq y] \cdot \mathbb{P}[Y_2 \leq y] \cdots \mathbb{P}[Y_n \leq y]$ (i.d.)

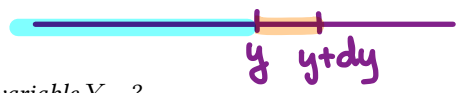
$= (\mathbb{P}[Y_1 \leq y])^n = (F_Y(y))^n$

Problem 13.3. *Assume that the random sample comes from a density f_Y. Is the r.v. $Y_{(n)}$ continuous? If so, what is its density $g_{(n)}$?*

$\longrightarrow$: For all y where F_Y is differentiable:

$$g_{(n)}(y) = \frac{d}{dy} F_{(n)}(y) = \frac{d}{dy}\left((F_Y(y))^n\right) =$$

$$= n \cdot (F_Y(y))^{n-1} \cdot f_Y(y)$$



Problem 13.4. *What is the distribution function of the random variable $Y_{(1)}$?*

$\longrightarrow$: for $y \in \mathbb{R}$: $F_{(1)}(y) = \mathbb{P}[Y_{(1)} \leq y] = \mathbb{P}[\min(Y_1, \dots, Y_n) \leq y]$

$= 1 - \mathbb{P}[\min(Y_1, \dots, Y_n) > y]$

$= 1 - \mathbb{P}[Y_1 > y, Y_2 > y, \dots, Y_n > y]$ (independence)

$= 1 - \mathbb{P}[Y_1 > y] \cdot \mathbb{P}[Y_2 > y] \cdots \mathbb{P}[Y_n > y]$ (i.d.)

$= 1 - (\mathbb{P}[Y > y])^n = \underline{1 - (1 - F_Y(y))^n}$

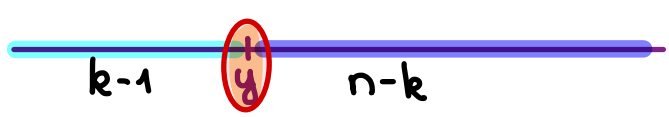
Problem 13.5. *Assume that the random sample comes from a density f_Y. Is the r.v. $Y_{(1)}$ continuous? If so, what is its density $g_{(1)}$?*

$\longrightarrow$: for y where F_Y is differentiable:

$$g_{(1)}(y) = \frac{d}{dy} F_{(1)}(y) = \frac{d}{dy}\left(1 - (\boxed{1 - F_Y(y)})^n\right)$$

$$= +n \cdot (1 - F_Y(y))^{n-1} (+1) \cdot f_Y(y)$$

$$g_{(k)}(y) = n \cdot \binom{n-1}{k-1} F_Y(y)^{k-1} f_Y(y) \, (1 - F_Y(y))^{n-k}$$



Theorem 13.2. *Lt $Y_1, \ldots, Y_n$ be independent, identically distributed random variables with the common cumulative distribution function F_Y and the common probability density function f_Y. Let $Y_{(k)}$ denote the kth order statistic and let $g_{(k)}$ denote its probability density function. Then,*

$$g_{(k)}(y) = \frac{n!}{(k-1)!(n-k)!}(F_Y(y))^{k-1} f_Y(y)(1 - F_Y(y))^{n-k} \quad \textit{for all } y \in \mathbb{R}.$$